

# CITATIONFLOW ANALYTICS

Research Citation Network Analysis and Influence Ranking System

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# ABSTRACT

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- CitationFlow Analytics analyzes relationships between research papers.
  - Each research paper is represented as a node.
  - A citation between two papers is represented as a directed edge.
  - The system builds a citation network to study the flow of knowledge.
  - It identifies influential research papers based on citation relationships.
  - It also visualizes citation paths and generates analytical reports.
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# PROBLEM STATEMENT

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- There are a large number of research papers published every year.
- Understanding the relationships between papers manually is difficult.
- It is hard to identify influential research papers in a large citation network.
- Tracking how one research paper influences other papers can be complex.
- A system is needed to efficiently analyze citation relationships and knowledge flow.

# PROPOSED FUNCTIONALITIES

The system will provide:

- Add and manage research papers
- Create citation relationships between papers
- Build a directed citation graph
- Identify influential research papers
- Analyze citation paths
- Calculate citation flow and paper rankings
- Visualize the citation network
- Generate citation analysis reports

# Tools and Technologies

- Java
- Directed Graph
- Adjacency List
- BFS
- DFS
- Edmonds-Karp (Maximum Flow)
- Priority Queue / Sorting
- Java Collections Framework
- HTML
- CSS
- JavaScript
- Java HttpServer

# Expected Outcome

- Create a network of research papers and citations
- Represent citation relationships using a directed graph.
- Identify important and influential research papers.
- Show how knowledge flows through the citation network.
- Display citation paths and relationships visually.
- Generate rankings and analytical reports.
- Efficiently handle multiple research papers and citation relationships.

# PROJECT IMPLEMENTATION:



8

Research papers (V)

15

Citation edges (E)

P1

Top paper by in-degree

3

Demo maximum flow

$O(VE^2)$

Edmonds-Karp

### 01 • MODEL

#### Directed Citation Graph

A research paper is a vertex. A citation is a directed edge from the **citing** paper to the **referenced** paper. Every edge has a positive capacity for the network-flow model.



ImageNet Classification with Deep Convolutional Neural Networks → ImageNet: A large-scale hierarchical image database  
capacity = 1



Deep Residual Learning for Image Recognition → ImageNet Classification with Deep Convolutional Neural Networks  
capacity = 1



Deep Residual Learning for Image Recognition → Very Deep Convolutional Networks for Large-Scale Image Recognition  
capacity = 1

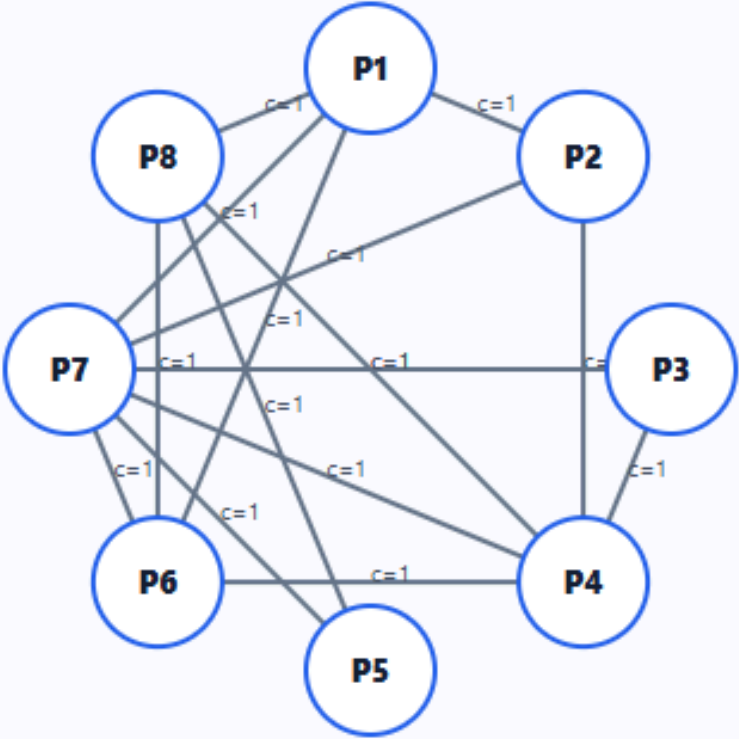


An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale → Deep Residual Learning for Image Recognition

02 • VISUALIZATION

Citation Network Map

The graph view makes the relationship structure easy to explain during the viva.



03 • DATA

Research Paper Dataset

The seeded records are real publications. The project uses a selected citation network for demonstration; it is not intended to reproduce the entire academic citation database.

| ID | Research paper                                                  | Authors                                                  | Year | Manage |
|----|-----------------------------------------------------------------|----------------------------------------------------------|------|--------|
| P1 | ImageNet: A large-scale hierarchical image database             | J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei | 2009 | Delete |
| P2 | ImageNet Classification with Deep Convolutional Neural Networks | A. Krizhevsky, I. Sutskever, G. E. Hinton                | 2012 | Delete |

#### 04 • TRAVERSAL

### BFS & DFS

BFS explores level-by-level using a queue. DFS explores deeply before backtracking. Both traverse the adjacency list in  $O(V + E)$ .

Start paper

P7 — Swin Transformer: Hierarchical Vision Transformer✓

Run BFS

Run DFS

**BFS order:** P7 → P1 → P2 → P3 → P4 → P5 → P6

#### 05 • PATH ANALYSIS

### Citation Path

Find a directed path from one paper to another using BFS and parent tracking.

Source / citing paper

P7 — Swin Transformer: Hierarchical Vision Transformer✓

Target / referenced paper

P1 — ImageNet: A large-scale hierarchical image database✓

Find Path

**Path:** P7 → P1

Edges traversed: 1

#### 06 • NETWORK FLOW

### Maximum Citation Flow

Uses **Edmonds-Karp**, a Ford-Fulkerson implementation that uses BFS to find augmenting paths in the residual graph.

Source

P7 — Swin Transformer: Hierarchical Vision Transformer✓

Target

P1 — ImageNet: A large-scale hierarchical image database✓

Calculate Maximum Flow

3

**Maximum modeled flow: P7 → P1**

1. P7 → P1 bottleneck 1
2. P7 → P2 → P1 bottleneck 1
3. P7 → P6 → P1 bottleneck 1

Capacity is a project-defined flow unit, not a Google Scholar citation count.

07 • RANKING

Influential Papers

The project ranks papers by **in-degree**, i.e. the number of incoming citation edges inside this selected network. A Java **PriorityQueue** produces the ranking.

| Rank | ID | Paper                                                                      | Incoming citations |
|------|----|----------------------------------------------------------------------------|--------------------|
| #1   | P1 | ImageNet: A large-scale hierarchical image database                        | 4                  |
| #2   | P4 | Deep Residual Learning for Image Recognition                               | 3                  |
| #3   | P2 | ImageNet Classification with Deep Convolutional Neural Networks            | 2                  |
| #4   | P3 | Very Deep Convolutional Networks for Large-Scale Image Recognition         | 2                  |
| #5   | P5 | Attention Is All You Need                                                  | 2                  |
| #6   | P6 | An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale | 2                  |
| #7   | P7 | Swin Transformer: Hierarchical Vision Transformer using Shifted Windows    | 0                  |
| #8   | P8 | Learning Transferable Visual Models From Natural Language Supervision      | 0                  |

08 • MANAGEMENT

Add Research Paper

Adds a new graph vertex to the current Java session.

**Demo:** use P9, then create P9 → P7 in the citation form.

ID

P9

Title

09 • RELATIONSHIP

Create Citation

**From** = citing paper → **To** = referenced paper. Capacity must be a positive integer.

From

P8 — Learning Transferable Visual Models From Natural Language Supervision



To

P7 — Swin Transformer: Hierarchical Vision Transformer using Shifted Windows



## 10 • COMPLEXITY

### Algorithm & Scalability Analysis

These complexity values directly support the computational-requirements discussion in the DSA PPT.

#### BFS

$O(V + E)$

Queue + visited set

#### DFS

$O(V + E)$

Recursion + visited set

#### Path BFS

$O(V + E)$

Parent reconstruction

#### Ranking

$O(E + V \log V)$

Indegree + PriorityQueue

#### Edmonds-Karp

$O(VE^2)$

Residual BFS iterations

## 11 • REPORT

### Generated Citation Analysis Report

**Current network:** 8 papers, 15 directed citation edges.

**Top in-degree paper:** P1 — ImageNet: A large-scale hierarchical image database (4 incoming edges).

**Demo flow:**  $P7 \rightarrow P1 = 3$  modeled flow units.

**Algorithms:** Adjacency List • BFS • DFS • BFS Path • PriorityQueue Ranking • Edmonds-Karp Maximum Flow.

**Complexities:** BFS/DFS/Path  $O(V+E)$ ; Ranking  $O(E+V \log V)$ ; Edmonds-Karp  $O(VE^2)$ .

Interpretation: this system demonstrates algorithmic citation-network analysis on a selected real-paper dataset. Edge capacity and in-degree are project metrics, not authoritative academic impact metrics.

[Download Report \(.txt\)](#)

# FUTURE SCOPE

- Real-time research paper data – Connect the system with research databases to automatically obtain updated papers and citation relationships.
- Large-scale citation analysis – Extend the system to handle thousands or millions of research papers efficiently.
- Advanced ranking algorithms – Implement algorithms such as PageRank to identify influential research papers more effectively.
- AI-based research insights – Use AI/ML to identify research trends and recommend related papers.
- Interactive visualization and cloud deployment – Provide advanced graph visualization and deploy the system as an online web application.

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# THANK YOU

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